

Asymmetric networks of beliefs

Echo talk

Henry Zwart

MSc. Computational Science

2026-02-17

Reminder of research goals

- Identify climate-change belief systems: How do beliefs and attitudes relate/reinforce, and how does this vary across population groups?
- Measure (simulated) implications of belief system structure on behavioural intervention effectiveness and outcome.
- Validate findings regarding effective intervention against related experimental work, and classical theories of belief change.

So far . . .

1. Data preparation
2. Initial exploratory data analysis

Data validation

Longitudinal data comprises six waves with variable participation.

Over 600 questions, which can vary with:

1. Different waves,
2. New and repeating participants,
3. Responses to other questions,
4. Experimental treatment.

Validation required to ensure that:

1. Questions are displayed exactly when we expect, and
2. All responses fit the expected format.

Data validation: Stage 1 (Schema)

Check that the expected columns are present with the correct data type, and all non-null response values are valid.

Pandera schema does most of the heavy lifting.

```
1 class ResponseSchema(BaseSchema):
2     response_id: pl.UInt32
3     wave: int = pa.Field(isin=WAVES)
4     participant_id: pl.UInt32 # Columns are non-nullable by default
5     dem_age: int = pa.Field(gt=0, le=99) # Age limits as specified in codebook
6
7     cc1: int = pa.Field(isin=[0, 1, 99], nullable=True) # Q: Climate change is happening (T/F/?)
8     cc_scam: int = pa.Field(isin=[1, 2, 3, 4, 5], nullable=True) # Q: Climate change is a scam
9     ...
10
11 ResponseSchema.validate(response_df) # Falls over if any schema req. not met
```

 Python

Data validation: Stage 2 (null response $\iff$ null expected)

Question responses should be non-null iff

- Question is shown (in wave, to participant type), and
- Any conditional-display logic is satisfied.

Three checks:

Not asked in wave $\implies$ response is null

Conditional logic not satisfied $\implies$ response is null

Question is displayed $\implies$ response is not null

Identifies 33 errors among 163 questions currently considered.

Additional data cleaning

Null ID participants: Removed from dataset since cannot be linked across waves.

Treatment columns:

- Coalesced treatment-specific responses into single column, and treatment indicators into an index column.
- Added variant columns for real-valued treatments (e.g. policy costs).

Before

participant_id	ccSolve0	ccSolve1	ccSolve10	ccSolve50	ccSolve100	WTP0	WTP1	WTP10	WTP50	WTP100
i64	i32	i32	i32	i32	i32	i32	i32	i32	i32	i32
1067064865	null	null	null	2	null	null	null	null	1	null
1073552977	null	5	null	null	null	null	1	null	null	null
1073965853	5	null	null	null	null	1	null	null	null	null
1074860181	null	null	null	null	2	null	null	null	null	1
1079191581	null	null	null	3	null	null	null	null	1	null

After

participant_id	ccSolving	Group_ccSolving	Variant_ccSolving
u32	i64	i64	i64
1067064865	2	3	50
1073552977	5	1	1
1073965853	5	0	0
1074860181	2	4	100
1079191581	3	3	50

Additional data cleaning

Categorical columns: Converted integer-valued responses to enums:

- 1. Human-readability, and
- 2. Consistent category values across waves.

Before

participant_id	ew1
i64	str
1067064865	"4"
1073552977	"2"
1073564910	"4"
1073965853	"2,4,8"
1074860181	"2"

After

participant_id	ew1
u32	list[enum]
1067064865	["Heat wave"]
1073552977	["Hurricane"]
1073564910	["Heat wave"]
1073965853	["Hurricane", "Heat wave", "Earthquake"]
1074860181	["Hurricane"]

Mapping survey questions to cognitive states

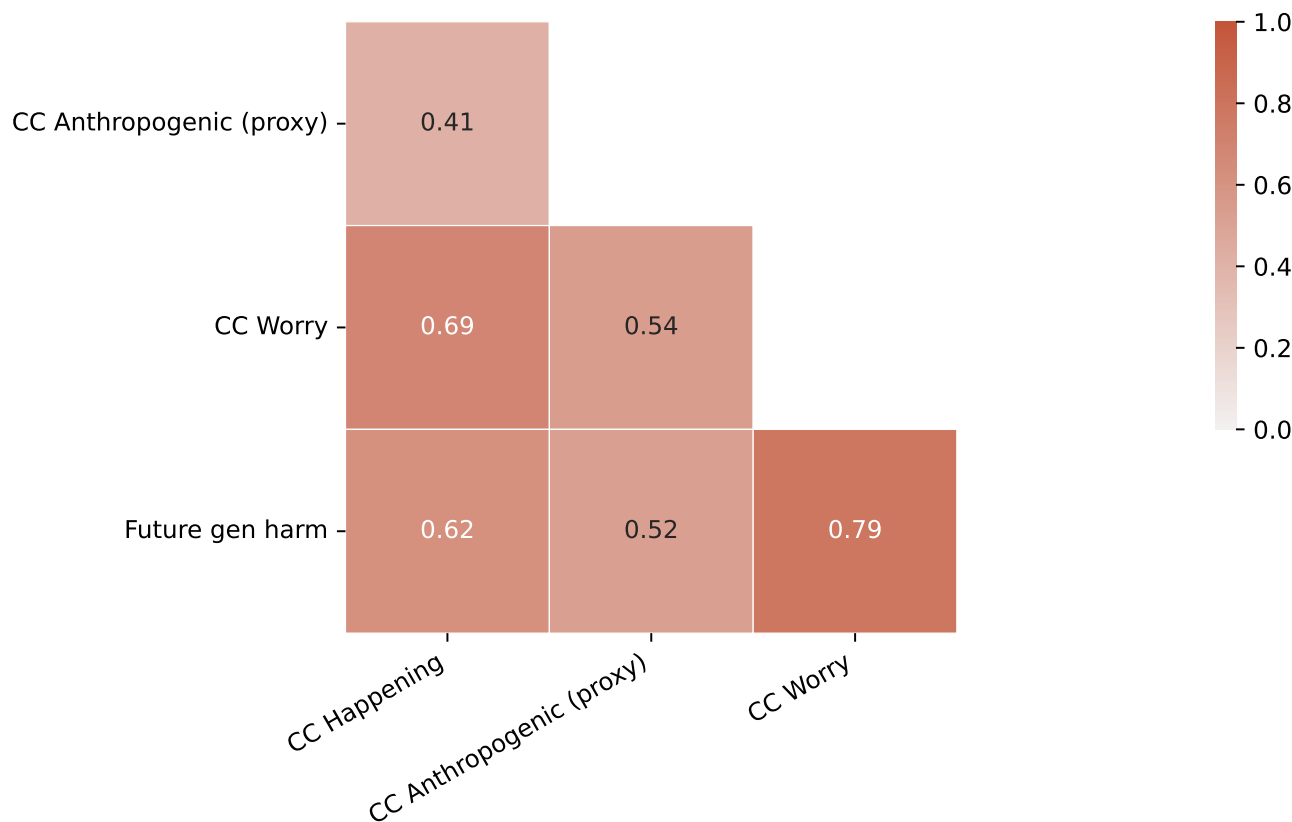
Working from S. Lee *et al.* (2025) as basis paper.

Eight cognitive states:

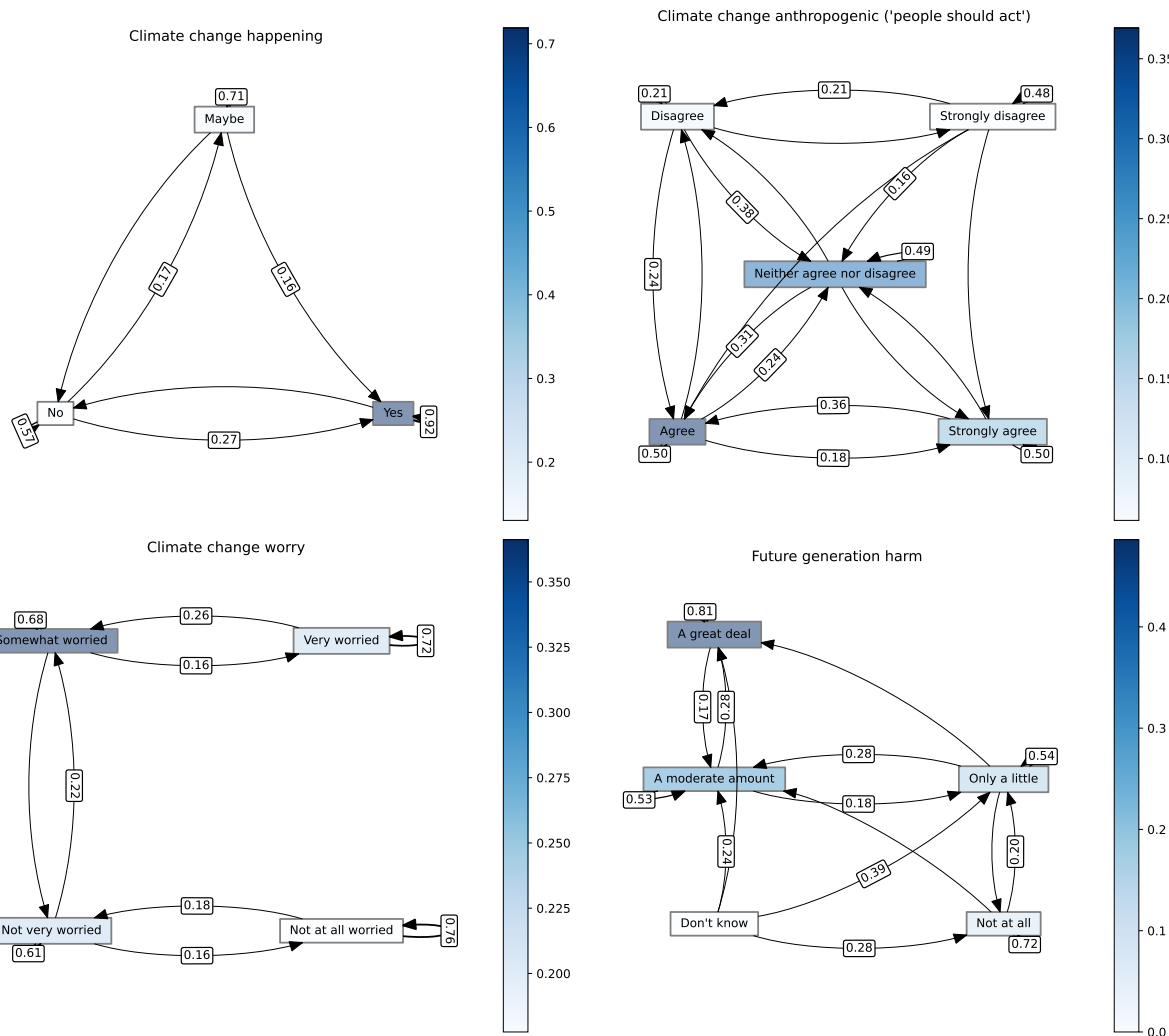
Type	State	Well-represented
Belief	Climate change is happening	✓
	Climate change is anthropogenic	✓
Risk perception	Climate change worry	✓
	Climate change personal harm	
	Climate change future gen. harm	✓
Policy support	Fossil fuel reduction	
	Renewable energy increase	
	Govt. priority for climate change	

Criteria: Assessed in multiple waves, positive/negative responses both meaningful.

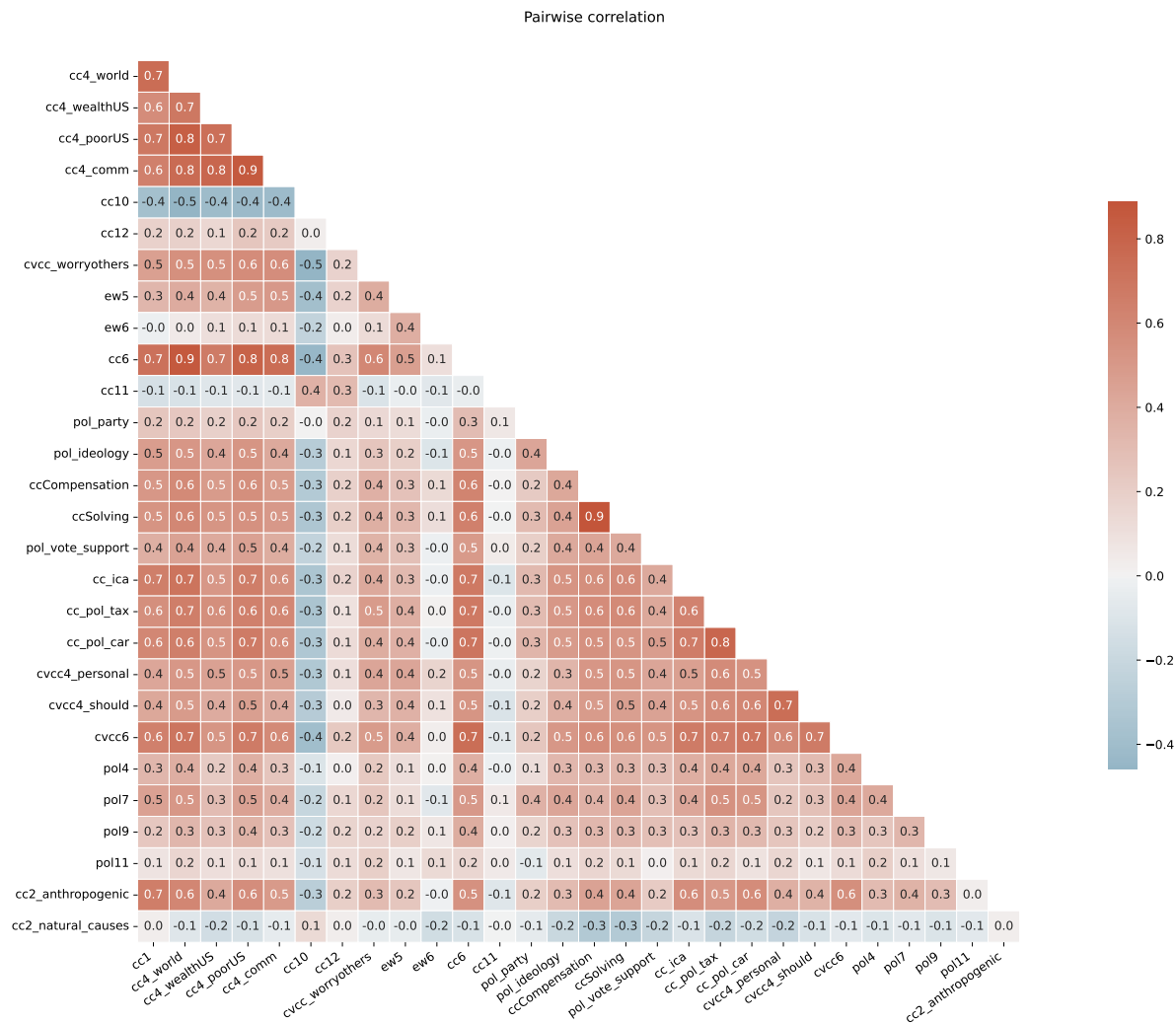
Lee 2025 representative questions: Pairwise correlation



Lee 2025 representative questions: Response transitions



Extended question set



Bibliography

- [1] S. Lee *et al.*, “Variations in Climate Change Belief Systems across 110 Geographic Areas,” *Nature Climate Change*, vol. 15, no. 9, pp. 947–953, Sept. 2025, doi: 10.1038/s41558-025-02410-1.